MIL-S-80274
7 July 1987
SUPERSEDING
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27 October 19'76

MILITARY SPECIFICATION

SAWS, POWER, HACK, 6-INCH THROUGH 24-INCH WORKPIECE CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manual and automatic bar feed hack saws for cutting off bars, rods, pipe, tubing and structural shapes of ferrous and non-ferrous metals.

1.2 Classification. The hacksaws shall be of the following styles and sizes. The style and size supplied shall be as specified (see 3.6 and 6.2.1).

Style I - Manual bar stock feed, non-automatic cutting

Style II - Automatic bar stock feed, automatic cutting

Size 6 - 6-inch width by 6-inch height
Size 10 - 10-inch width by 10-inch height
Size 12 - 12-inch width by 12-inch height
Size 14 - 14-inch width by 14-inch height
Size 18 - 18-inch width by 18-inch height
Size 20 - 20-inch width by 20-inch height
Size 24 - 24-inch width by 24-inch height

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B11.10 - Safety Requirements for the Construction, Care, and Use of Metal Sawing Machines.

ANS1/NEMA MG 1 - Motors and Generators.

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURER$ ASSOCIATION (AGMA)

AGMA 360.02 - Manual for' Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1698.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10118-3308.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

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3.2 Design. The hack saw shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The hack saw shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.I Measurement system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the hack saw. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The hack saw may contain reclaimed materials provided such materials will not jeopardize the hack saw's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the hack saw.

3.2.3 Energy efficiency. The hack saw and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the hack saw that present safety hazards. The safety devices shall not interfere with the operation of the hack saw. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All hack saw parts, components, mechanisms, and assemblies furnished on the hack saw, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 that are applicable to the hack saw itself. In addition, the machine shall comply with all the requirements of ANSI B11.1O that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of OSHA and ANSI standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The hack saw shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the hack saw.

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3.2.8 Environmental protection. The hack saw shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the hack saw shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The hack saw shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the hack saw to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux , and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the hack saw shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts used on the hack saw and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

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3.3.7 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be proper width and size to transmit full rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. Saw covered herein shall consist essentially of a base, saw blade frame, reciprocating drive mechanism, feed mechanism, work holding vise and a coolant system. In addition, the style 11 hack saws shall have an automatic bar feed unit.

3.4.1 Base. The base shall support the saw frame, drive mechanism, feed mechanism, workholding vise and such other components mounted thereon. The base shall have sufficient strength and rigidity to support the working parts of the saw and a workpiece equalling the maximum capacity of the saw. Rigidity shall be such that alignment of all parts is maintained to the degree necessary to produce cutting rates and accuracies specified herein. The base shall have access openings for maintaining internal mechanisms and removal of chips. The base shall have means of leveling and securing the saw Lo the floor or foundation. When specified (see 6.2.1), the base of size 18 by 18 and 24 by 24 saws shall have such features as necessary to position the saw blade for producing cuts of not less than 45 degrees through the workpiece axis. Either the base shall be constructed to swivel or columns supporting the blade drive and feed mechanism shall be adjustable, horizontally. When this feature is furnished, a non-swivel vise (3.4) shall be furnished on the saw. A degree scale shall provide cutting any angel through 45 degrees to the workpiece axis.

3.4.2 Frame. A frame supporting the saw blade under tension shall reciprocate either on anti-friction bearing or on gibbed ways. All parts in the reciprocating frame shall either be adjustable or replaceable to maintain working tolerances and necessary alignment of parts. The frame shall function in such a manner that at the end of each cut, the frame shall return to, and remain in, the elevated or retracted position for loading the next workpiece.

3.4.3 Blade drive mechanism. The blade drive mechanism shall consist of an electric motor operating in conjunction with a belt, transmission device and chain drive system, or combinations thereof. The blade drive mechanism shall provide reciprocal blade strokes per minute, either through a number of speed selections, or a speed range which is infinitely variable, to meet the requirements of table 1. When specified (see 6.2.1), either a single blade speed, multiple blade speeds or infinitely variable blade speeds shall be furnished as required.

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3.4.4 Blade feed mechanism. The blade feed mechanism shall be either mechanical, hydraulic or combination mechanical and hydraulic. The blade feed system shall have either manual adjustments or controls to provide efficient, automatic cutting pressures required for various sizes, shapes and materials to be cut. The blade feed system shall operate in conjunction with the reciprocating action of the saw blade, automatically applying pressure on the cutting stroke and lifting the blade on the return non-cutting stroke, while maintaining positive feed rates through entire cut off cycles.

3.4.5 Worktable and vise. The worktable and vise shall have a strength and rigidity to support and hold a ferrous workpiece of a size equal to the maximum rated capacity, required by table I, while producing a metal removal rate required by table II. Unless otherwise specified in 3.4.1.6, the worktable shall be so constructed that coolant will gravity flow back to the sump or reservoir. The vise shall be of the quick acting type. Unless otherwise specified (see 6.2.1), style I saws shall have a vise of the swivel type and Style II saws shall have a non-swivel type vise. The swivel vise shall be capable of being positioned to cut a workpiece at any angle from 90 degrees to the workpiece axis.

3.4.6 Bar feed unit. Style II saws shall have a bar feed unit which shall feed the workpiece in increments over a range of 1/4 inch or less, to 24 inches or more, with a repeatable accuracy + 0.010. The bar feed unit shall consist essentially of powered vises and a feeding mechanism. The vises shall have means of applying clamping pressure on thin wall tubing, without distortion or other damage to the tubing; and also apply adequate pressure to securely retain workplaces which are solid and of maximum size within the capacity of the saw. The vises may be operated either electrically, hydraulically, mechanically or combinations thereof. When specified (see 6.2.1), the bar feed unit shall have a cut counter for selecting the number cuts to be made. The counter shall have a mechanism that will stop the saw when the selected number of cuts are completed. The bar feed unit shall be automatic after manual actuation of the cycle start button. The bar feeder, conveyor and vise assembly shall feed the workpiece to the preselected length, the vise shall clamp the workpiece and the saw blade shall cut the workpiece. Upon completion of the cut, the head shall retract and the above cycle shall be repeated until the last cut, after which the saw shall stop. The bar feed unit shall also stop the saw, in the event the workpiece is consumed, prior to completion of the selected number of cuts.

3.4.7 Loading and discharge tracks. Style II hack saws shall be provided with workpiece loading and discharge tracks. Unless otherwise specified (see 6.2.1), the loading track shall be 12 feet in length or longer, and the discharge track shall be 42 inches in length or longer. The tracks shall be so constructed that cut workplaces will not fall back into the saw blade.

3.4.8 Workpiece support stands. When specified (see 6.2.1), each Style I hack saw shall be furnished with three stock support stands. The stands shall be of the roller type capable of accommodating a workpiece of the maximum rated size of the saw. The stands shall incorporate means for height adjustment.

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3.4.9 Hydraulic system. Hydraulic systems, when furnished, shall be complete with pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in pressure lines and on the low side of each pressure reducing valve. Means shall be provided to determine and maintain fluid level.

3.4.10 Coolant system. Unless otherwise specified (see 6.2.1), each saw shall be equipped with a coolant system. The coolant system shall consist essentially of a sump or reservoir, a power driven pump, coolant control, a removable strainer and necessary piping. The sump or reservoir shall have a capacity to provide continuous flow of coolant and have means for filling, draining and cleaning. Means shall be provided to observe the coolant level within the reservoir or sump.

3.5 Electrical. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each hack saw shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hz circuit. Each hack saw shall have a fused safety disconnect switch or circuit breaker. The hack saw shall be initially wired for operation on 460 volts. An identified terminal for grounding the hack saw when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.5.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60 Hz. Unless otherwise specified (see 6.2.1), each motor shall have meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MC-1.

3.5.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.5.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.6 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the hack saw shall be not less than those specified in table I for the hack saw size specified.

3.7 Performance. The saw shall be capable of cutting workpieces, which are flat and perpendicular to the workpiece axis within + 0.002 inch per inch, at the cutting rates listed in Lable II.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the in equipment shall be shall be furnished with each saw:

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a. One set of hand tools normally furnished by the manufacturer.
b. Chart or data plate indicating recommended speeds and feeds for general hack sawing operations.
c. Worklight.
d. Stock stop gage.
e. Saw blades, high-speed stel edge; 6 each, 6-teeth per inch and 6 each 10-teeth per inch.

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.10 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a corrosion-resistant lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each hack saw and contain the information listed below. If" the hack saw is a special model, the model designation shall include the model of the basic standard hack saw and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the hack saw shall be commensurate With all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The hack saw shall be examined to determine compliance with all requirements of this specification including compliance with all referenced standards.

4.7 Tests.

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4.7.1 Operational test. The hack saw shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each hack saw shall be tested to determine that it complies with the alignment tolerance requirements of 3.6.

4.7.3 Performance tests. The type saw blade used in the sawing test shall be at the option of the supplier. The workpiece shall have a nominal dimension equal to the rated capacity of the saw. Either a round or rectangular shaped workpiece may be used in the sawing test. The saw blade selected by the supplier shall make three cuts or more through a SAE 1045 cold rolled steel workpiece. The metal removal rate, flatness and perpendicularity shall conform to 3.6 and table II.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.2.3.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Hack saws covered by this specification are intended for use on various ferrous and non-ferrous materials.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size required (see 1.2 and 3.6).

c. First article, if required (see 3.1).

d. If hack saw design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. Additional safety and health requirements, if required (see 3.2.5).

f. Painting, if different (see 3.3.5).

g. If electromagnetic interference control is required,

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specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

h. Base features for angular cuts, if required (see 3.4.1).

i. Blade speeds, if different (see 3.4.3).

Jo Describe type vise, if different (see 3.4.5).

k. Cut counter if required (see 3.4.6).

l. Describe loading and discharge requirements for style 11 saws, if different (see 3.4.7).

m. Workpiece support stands, if required (see 3.4.8).

n. Delete coolant system, if not required (see 3.4.10).

o. Electrical system and input power, if different (see 3.5).

P. Motor enclosure, if different (see 3.5.1).

q. Size and capacity, if different (see 3.6).

r. Standard equipment, if different (see 3.8).

s. Specify and fully describe optional equipment, if required (see 3.9).

t. Specify applicable repair parts, if required (see 3.10).

u. Lubrication chart or plate, if different (see 3.11.1).

v. Nameplate, if different (see 3.11.2).

w. Inspection condition, if different (see 4.3).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (see 4.5).

z. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the-contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the

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first article.

6.4 Subject term (key word) listing

Bar feed, automatic, non-automatic
Bar, pipe, rod, tube cutting
Saws, power hack

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I - Sizes and capacities.

Characteristics		Size 6	Size 10	Size 12	Size 14	Size 18	Size 20	Size 24
Size, rectangular dimensions, workpiece cross section, inches								
Normal 90° cutoff		6 x 6	10 x 10	12 x 12	14 x 14	18 x 18	20 x 20	24 x 24
Angular 45° cutoff		4 x 6	6 x 10	7 x 12	9 x 14	12 x 18	14 x 20	15 x 24
Blade data:								
Length, inches	Standard	14	18	18	21	21	24	36
	Optional	---	17 - 19	21	18 - 24	30	30	30
Stroke, inches	Standard	5	5	6	6	4-1/2	6	8
	Optional	5-1/2	6	---	---	---	---	6
Stroke, per minute		80 - 160	80 - 160	50 - 145	50 - 145	35 - 145	35 - 145	25 - 145
Motor horsepower								
	Standard	1	3	5	5	3	5	5
	Optional	1-1/2-3	2 - 5	7-1/2	5-7-1/2	5-7-1/2	7-1/2	10

TABLE II - Performance.

	Size 6	Size 10	Size 12	Size 14	Size 18	Size 20	Size 24
Cutting rates (SAE 1045 cold rolled steel bar, square inches per minute)	4	5	7	7	5	5	5

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84

User activities:
Navy - MC, OS

Preparing activity:
DLA-IP

(Project 3405-0080)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-S-8027/4	Saws, Power, Hack, 6-Inch Through 24-Inch Workpiece Capacity

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE

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